

Ding Dong Touch Bell

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Ding dong bell is a popular nursery rhyme. Shakespeare used the phrase “ding dong bell” in several plays. Today, ding dong is a popular ringtone used in phones and door bells. Here is a simple ding dong tone generator circuit built around a dedicated analogue IC.

Circuit and working

Fig. 1 shows the circuit of ding dong tone generator. The circuit is built around BT8031 ding dong IC, CD4049 inverting buffer, op-amp 741, transistors BD678 and BC547, touch plate, speaker and a few discrete compo-

nents. It requires 9V DC supply.

BT8031 is a 3-pin ding dong IC in TO-92 package with operating voltage of 1.3-3.3V.

As seen from the circuit, the touch signal from the touch plate is picked up by op-amp IC 741 (IC1), and inverted and buffered by IC CD4049 (IC2). Output pin 6 of IC2 drives BT8031

Test Points

Test point	Details
TP0	0V, Ground
TP1	9V DC
TP2	2.2V when touched
TP3	High when not touched
TP4	High when not touched
TP5	High when touched



PARTS LIST

Semiconductors:

- IC1 - 741 op-amp
- IC2 - CD4049 inverting buffer
- IC3 - BT8031 ding dong IC
- T1 - BC547 npn transistor
- T2 - BD678 pnp transistor
- ZD1 - 2.2V zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R4 - 10-kilo-ohm
- R2, R3 - 15-kilo-ohm
- R5 - 22-ohm
- R6 - 560-ohm
- VR1 - 2-mega-ohm preset

Capacitors:

- C1 - 0.1 μ F ceramic
- C2 - 10 μ F, 25V electrolytic

Miscellaneous:

- LS1 - 8-ohm, 0.5W loudspeaker
- Touch plate
- 9V DC power supply

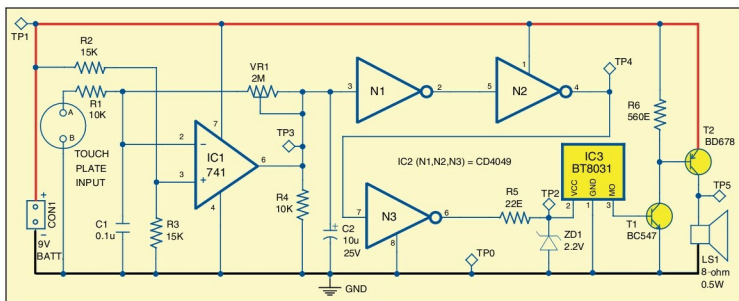


Fig. 1: Circuit of ding dong tone generator

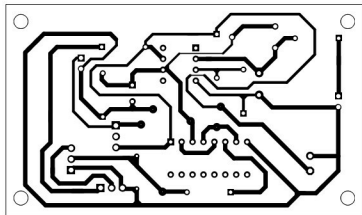


Fig. 2: An actual-size, single-side PCB for the ding dong tone generator

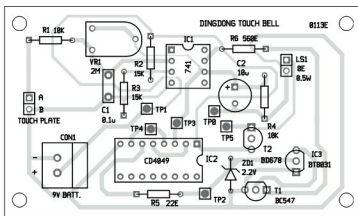


Fig. 3: Component layout for the PCB